Final Project Report - Stockholm Inside Airbnb Data Engineering Assessment

1. Executive Summary

This project builds a reproducible data engineering and analytics workflow for the Stockholm, Sweden Inside Airbnb dataset. The work covers ingestion, profiling, cleaning, validation, SQLite storage, exploratory data analysis, statistical testing, and a simple ML baseline for listing price prediction.

The most important dataset limitation is that Stockholm calendar `price` and `adjusted_price` are 100% missing in the raw source data. Therefore, calendar data is used only for availability analysis, while pricing analysis uses `listings_clean.price`.

2. Objectives and Scope

The objective is to demonstrate practical data engineering and analysis skills for a Data Engineer Intern technical assignment. The scope includes clean data pipelines, documented assumptions, business-focused EDA, simple statistical tests, and a beginner-friendly ML baseline. Production orchestration, dashboards, advanced ML, and external data are out of scope.

3. Dataset Overview

- Selected city: Stockholm, Sweden
- Data source: Inside Airbnb
- Cleaned listings: 4,955 rows
- Cleaned calendar rows: 1,808,575 rows
- Cleaned reviews: 161,036 rows
- Cleaned neighbourhoods: 14 rows
- SQLite database: `data/processed/airbnb_stockholm.db`

4. Data Engineering Approach

Day 1 loads raw CSV files from `data/raw`, profiles each dataset, cleans important fields, saves processed CSVs, creates SQLite tables, and writes a data quality report. The main command is:

```
python src/run_day1_pipeline.py
```

The project uses simple Python modules, `pathlib`, pandas, and SQLite. Raw and processed data files are ignored by Git.

5. Data Quality Findings

- Calendar `price` and `adjusted_price` are 100% missing.
- Listings `price` has 35.62% missing values.
- Price-based analysis excludes missing listing prices.
- Date, boolean, percentage, and money fields are cleaned where available.
- Invalid non-positive prices are flagged, not silently dropped.

6. EDA Findings

Day 2 EDA is generated by:

```
python src/run_day2_eda.py
```

Key findings:

- Median listing price is 1,200.00 in the source dataset currency.
- `Entire home/apt` is the largest room type with 4,066 listings.
- `Entire home/apt` also has the highest median room-type price at 1,386.00.
- Calendar availability is 38.21%.
- `Norrmalms` has the highest median price among the top 10 neighbourhoods by listing count.
- The top host by listing count is `Lukas` with 51 listings.

Each EDA chart includes both a finding and a business interpretation in `reports/eda_report.md`.

7. Statistical Findings

Day 3 statistical tests are generated by:

```
python src/run_day3_statistics.py
```

Summary:

- Entire homes and private rooms have different listing price distributions. Entire homes have a median price of 1,386.00, compared with 597.00 for private rooms. The Mann-Whitney U test p-value is < 0.001.
- Superhost and non-superhost review score comparison has p-value 0.7414, so this test does not show a statistically significant difference in score distributions.
- Listings with more than 10 reviews have a median price of 1,142.00, compared with 1,273.00 for listings with 10 or fewer reviews. The p-value is 0.0103.
- Neighbourhood price differences are statistically significant by Kruskal-Wallis test with p-value < 0.001.

Statistical significance does not prove causation. Business significance should be considered separately.

8. Optional ML Baseline Experiment

The ML baseline is generated by:

```
python src/run_day3_ml.py
```

Target: `listings_clean.price`.

Rows with missing price are excluded. Prices above the 99th percentile are removed for modeling only to reduce distortion; source data is not modified.

Model comparison:

Model	MAE	RMSE	R2
Gradient Boosting Regressor	462.98	685.49	0.5790
Random Forest Regressor	470.42	697.16	0.5646
Linear Regression	521.79	767.62	0.4721
Median price baseline	776.44	1,095.37	-0.0749

The best model by MAE is Gradient Boosting Regressor. This is only a baseline and is not production-ready.

9. Business Recommendations

- Segment pricing by room type before comparing listings.
- Use neighbourhood median prices as local benchmarks, while also considering room type, property type, and capacity.
- Use calendar data for availability analysis only.
- Interpret availability carefully because unavailable dates may be booked, blocked, or inactive.
- Use ML results as directional support, not automated pricing decisions.

10. Limitations

- Calendar prices are unusable because they are 100% missing.
- Listings with missing prices are excluded from price-based analysis.
- Availability is not the same as occupancy.
- Review count is not the same as booking count.
- Statistical tests show association, not causation.
- The ML model does not include external demand, events, competitor pricing, or true booking data.

11. Incomplete Work and Prioritization

Completed work prioritizes correctness, reproducibility, and clear interpretation. Remaining future work could include cross-validation, richer feature engineering, additional hypothesis tests, and a dashboard. These were not prioritized because the assignment benefits more from a clean end-to-end baseline than from over-engineered tooling.

I prioritized a clean, reproducible end-to-end workflow over adding advanced tools, because reliable data preparation and transparent assumptions are the foundation for later analytics and ML work.

12. Reproducibility Instructions

Download the Stockholm Inside Airbnb raw files and place them in `data/raw`:

- `listings.csv.gz`
- `calendar.csv.gz`
- `reviews.csv.gz`
- `neighbourhoods.csv`

Then run:

```
python src/run_day1_pipeline.py
python src/run_day2_eda.py
python src/run_day3_statistics.py
python src/run_day3_ml.py
```

Main reports:

- `reports/data_quality_report.md`
- `reports/eda_report.md`
- `reports/statistical_report.md`
- `reports/ml_experiment_report.md`
- `reports/final_report.md`
- `reports/ai_usage_disclosure.md`

13. AI Usage Disclosure Summary

AI tools used: ChatGPT and Codex. AI assisted with project structuring, code generation, report drafting, and explanation. Outputs were validated by running scripts locally against the actual Stockholm data and reviewing generated results. Full disclosure is available in `reports/ai_usage_disclosure.md`.